Ju Qi

Researcher in LLM, RL and Game Theory

Personal Homepage

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Education

- **Huazhong University of Science and Technology** — Wuhan, China
  Ph.D. in Control Science and Engineering — *Sep. 2019 - Mar. 2026*
- **Research area**: Large Language Model, Reinforcement Learning, Game Theory

Skills

- **Programming and Frameworks**: Proficient in Python, LaTeX, PyTorch; Experienced with C/C++, RLlib, OpenSpiel
- **Mathematics**: Game Theory, Mathematical Modeling, Reinforcement Learning, Learning Theory
- **Languages**: Chinese (native), English (fluent), Japanese (daily communication)

Experience

- **LLM Post-training Based on Games** — Individual Contributor
  ByteDance Seed — *Jun. 2025 - Oct. 2025*
  - **Project Goal**: Integrate game theory problems into LLM post-training to enhance LLM capabilities.
  - **Project Results**: Delivered game-oriented LLM evaluation & training frameworks. Trained an LLM Texas Hold'em AI that outperformed GPT-o3, Gork4, etc., with improved instruction-following.
  - **Personal Work**: Proposed a novel algorithm with "LLM Reflection" mechanism (outperforming traditional RL methods in game scenarios). Developed game-oriented LLM evaluation framework (integrated into team's system) and training framework based on Verl.
- **Texas Hold'em AI** — Project member, team of four
  Fen AI Lab (Remote) — *Sep. 2023 - Jan. 2024*
  - **Project Goal**: Created an AI that matches the performance of Pluribus (a renowned Texas Hold'em AI).
  - **Project Results**: The final AI reaching the level of professional players in two-player Texas Hold'em, and the multi-player Texas Hold'em AI is still under development.
  - **Personal Work**: Contributed to the development of key algorithms (e.g., MCCFR, MCCFR pruning), built foundational components such as strategy storage and result display, and handled algorithm parameter tuning and testing.
- **Reinforcement Learning Internship** — Main implementer, team of two
  ByteDance Nuverse — *Jul. 2021 - Mar. 2022*
  - **Project Goal**: Design multi-style AI companion NPCs for the game *One Piece: Burning Blood*
  - **Project Results**: Added a style evolution module to the previous AI training framework, leading to an 80–120% improvement in key indicators and clear style differentiation in play. Some AIs reached deployable quality.
  - **Personal Work**: I was the main executor of this project. Under my advisor's guidance, I implemented a multi-style AI algorithm and explored the integration of human preferences into reinforcement learning. This work culminated in a research paper summarizing the findings and potential applications.
- **Land Auction** — Project leader, team of six
  China Resources Group — *Feb. 2021 - Jun. 2021*
  - **Project Goal**: Design a bidding strategy for China Resources Group in the "first/last" land auction
  - **Project Results**: The strategy was approved by China Resources Land Group and deployed in dozens of land auctions (each over $100 million). Our algorithm outperformed the group's expert approach, boosting bid accuracy 3–4x and winning probability by 5%. The model was adopted as their standard land auction strategy.
  - **Personal Work**: Using past auction data, I created a simulation for "first/last" land auctions and applied the Fictitious Play algorithm to develop strategies. Participated in three actual land auctions, involving a total bidding scale of $1 billion, and refined the model based on real-world outcomes.

Publications

- **Preference-CFR Beyond Nash Equilibrium for Better Game Strategies (ICML 2025) [Link]**: Proposes the Preference Counterfactual Regret Minimization (Pref-CFR) algorithm to achieve diverse Nash equilibria, enabling customizable strategies by incorporating preference and vulnerability parameters. Demonstrates distinct play styles in Texas Hold'em without sacrificing strategic strength.
- **Accelerating Nash Equilibrium Convergence in Monte Carlo Settings Through Counterfactual Value Based Fictitious Play (NeurIPS 2024) [Link]**: Introduces the Monte Carlo Counterfactual Value-Based Fictitious Play (MCCFVFP) algorithm for large-scale games, achieving 20–50% faster convergence than standard MCCFR in complex settings like Texas Hold'em.
- **ELO-Rated Sequence Rewards: Advancing Reinforcement Learning Models [Link]**: Proposes ELO-Rated Sequence Rewards (ERRL), which uses ordinal preferences and ELO ratings to replace numerical rewards, achieving superior performance in long-term RL tasks like Atari benchmarks.

Personal Summary

Cheerful, optimistic, and thrives on tackling challenging tasks. Passionate about Go (WeiQi), gaming, and football. Currently seeking opportunities in LLM, RL, game theory, and multi-agent systems, aiming to advance both the frontiers and practical applications in these fields.